

Infosys Springboard Virtual Internship

Owner **Khrishwanth**

Instructor **Sai Vignesh S**

Modules **Authentication**

Problem statement

The insurance industry relies heavily on manual processes for creating, validating, and managing these policies, causing delays and errors. The task is to implement a system that can automate these tasks and improve the overall efficiency.

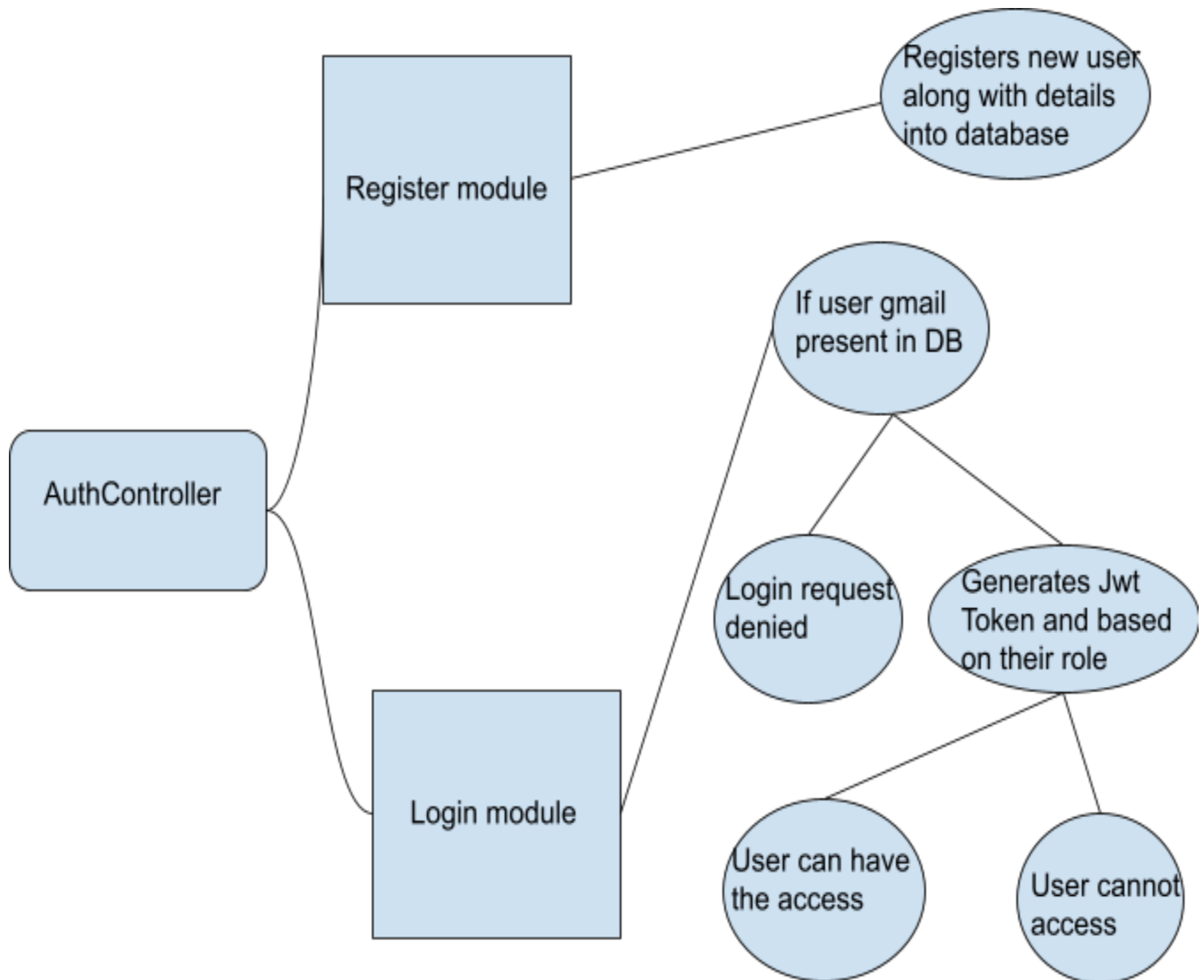
Objectives

Week	Status	Members	Plan
1	Completed ▾	Khrishwanth Ritvik	• Designed a module to authenticate and authorize users based on their roles
2	Completed ▾	Khrishwanth Ritvik Vedha	• Designed and Structured all Databases and its relationships to increase the performance
3	Completed ▾	Ritvik Khrishwanth Sindhu	• Designed the module for Agents and Users to create , view and update the appointment

My contribution for 4th & 5th one is brief since I don't know these concepts in detail. Yet I designed some functions in those modules . Rest of the members (**Nimisha**) contributed in designing other module like Notification & PolicyPlan Excluding these 5 modules

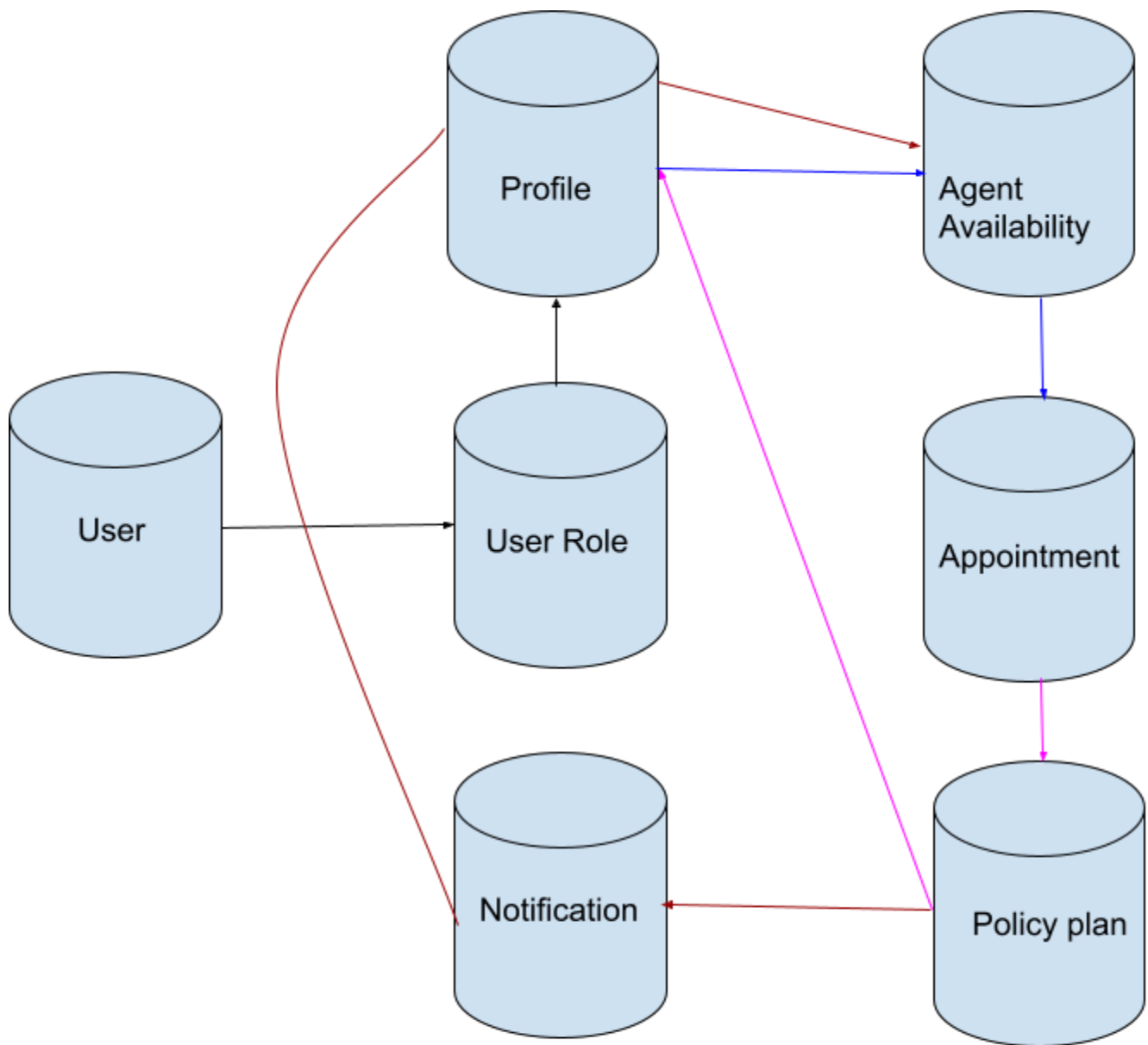
Nov 10, 2025

Module #1: Authentication



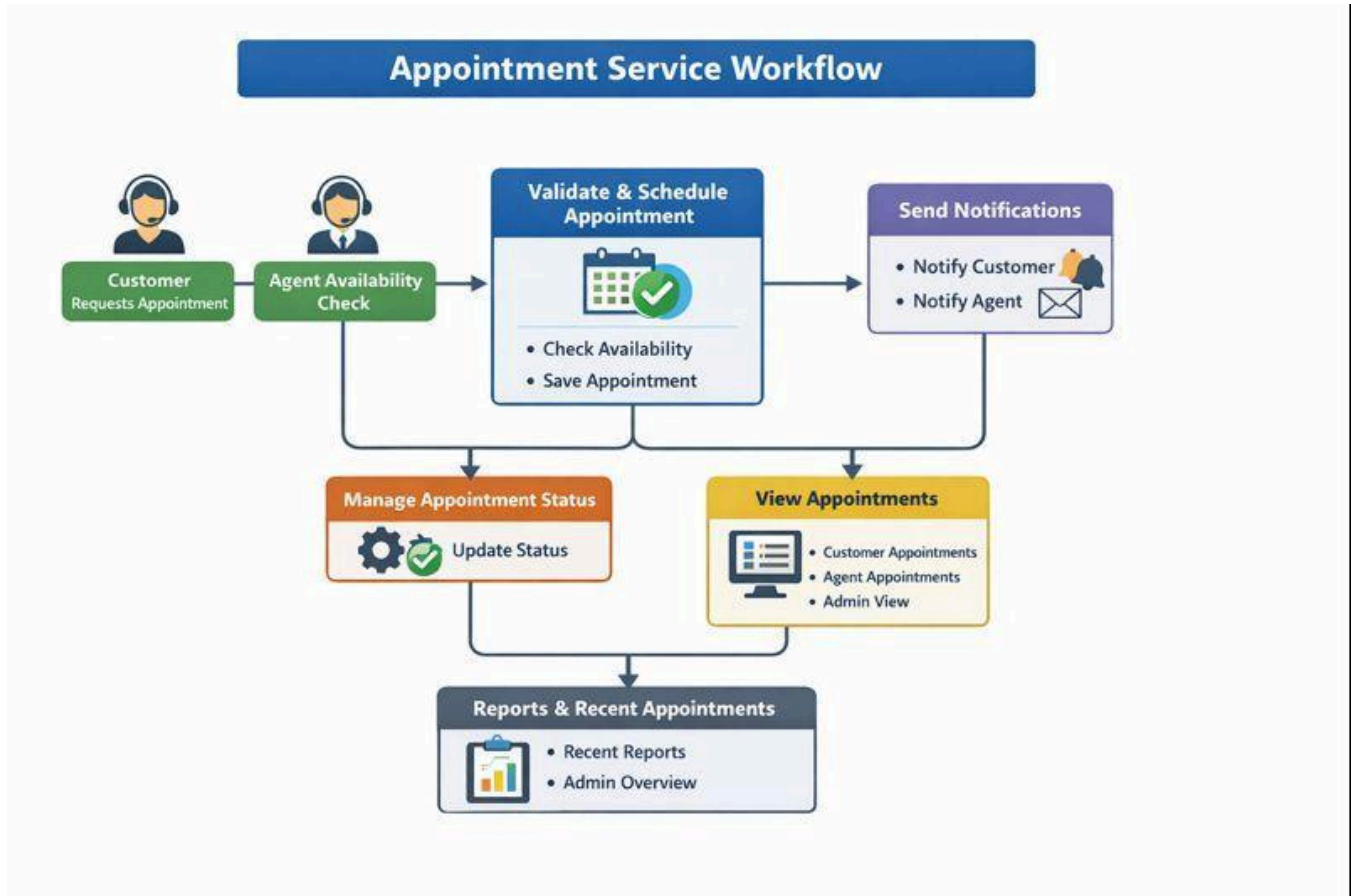
Nov 17, 2025

Module #2: DataBase Schemas



Nov 26, 2025

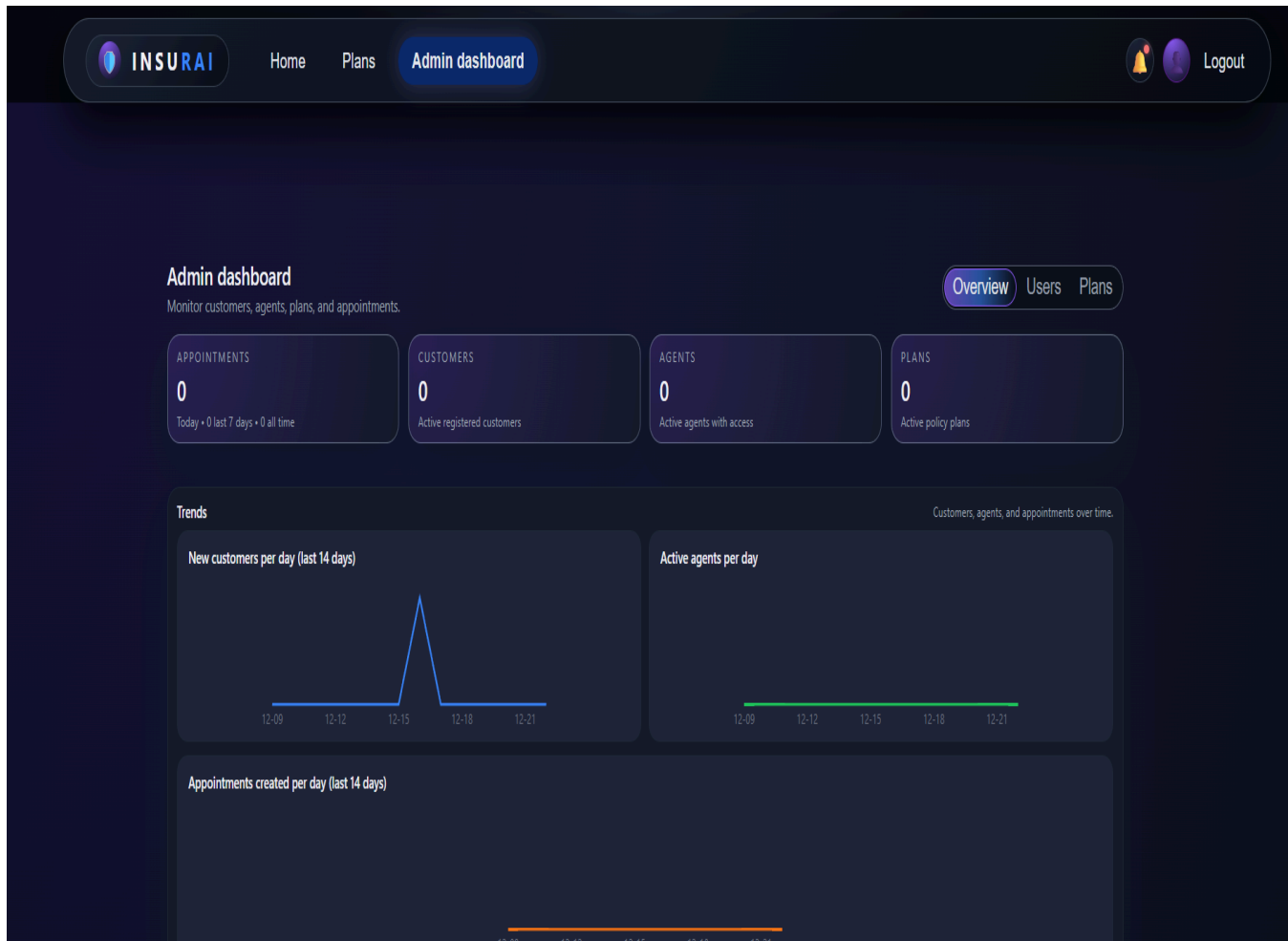
Module #3: Appointment scheduling



- Customer Request Appointment : **POST**
/api/appointments/schedule
- Customer View Appointments : **GET**
/api/appointments/customer/{customerId}
- Agent View Appointments : **GET**
/api/appointments/agent/{agentId}
- Agent Update Appointment Status : **PUT**
/api/appointments/{appointmentId}/status
- Admin View Recent Appointments : **GET**
/api/admin/appointments/recent

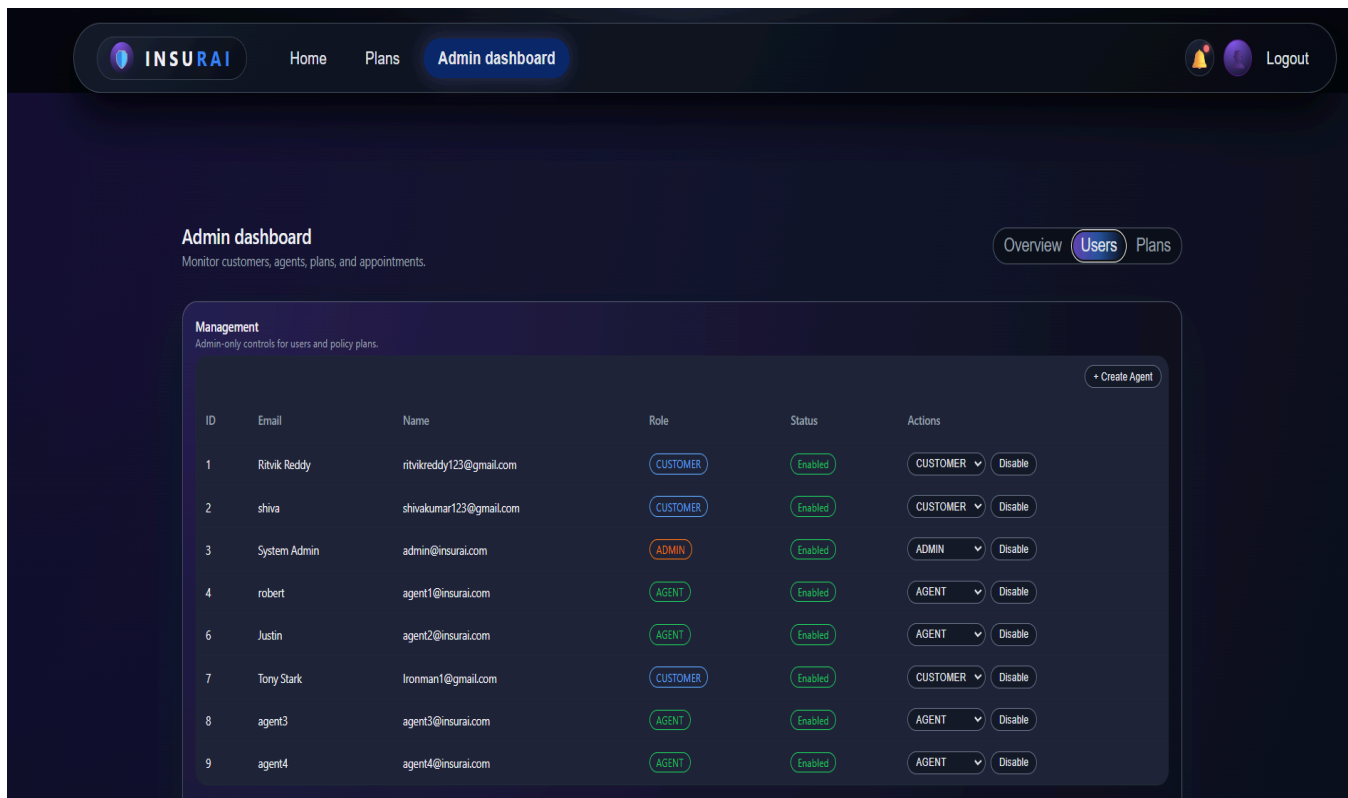
Dec 3, 2025

Module #4: Admin module



Finished and validated admin user management endpoints in AdminUserController:

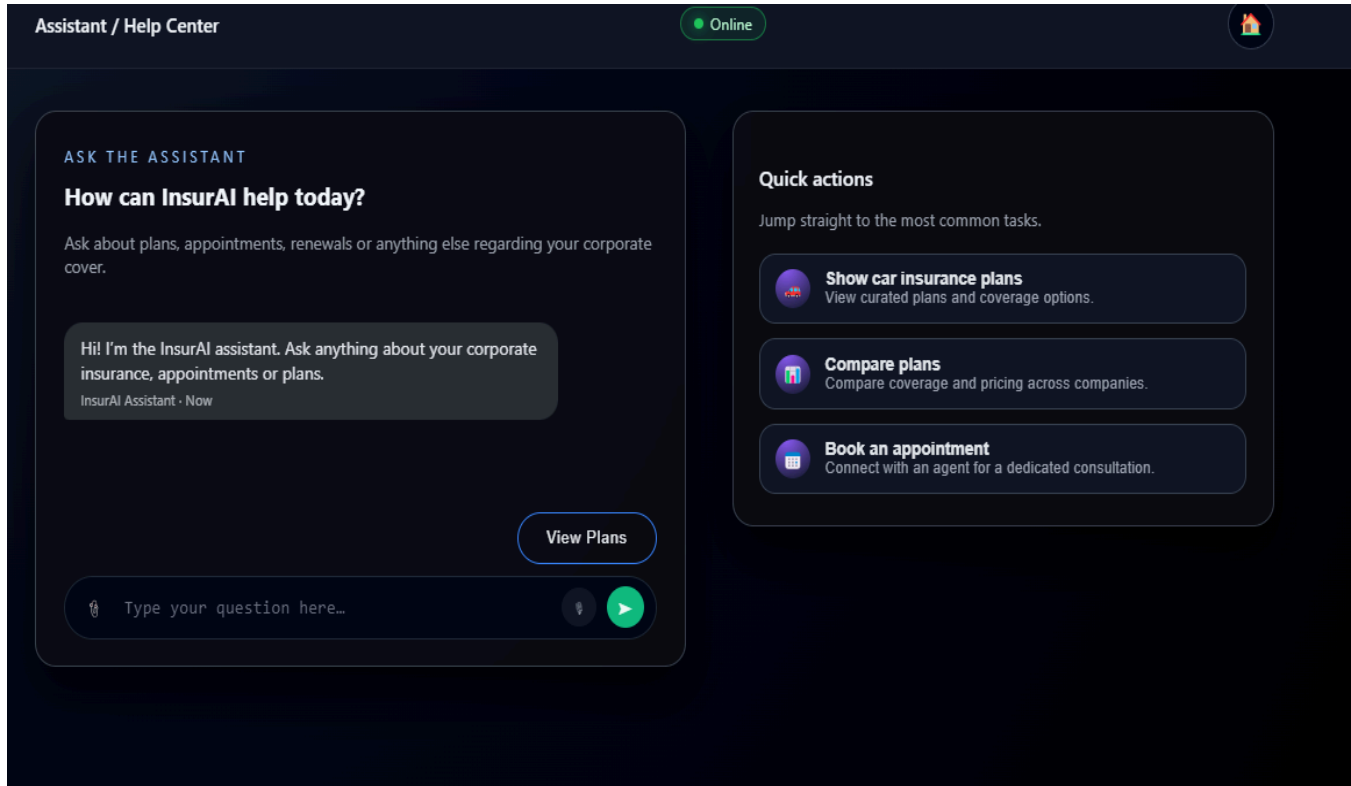
- **GET** /api/admin/users – list all users for AdminManagementPanel.
- **PUT** /api/admin/users/{id}/role and **PUT** /api/admin/users/{id}/status – update roles and active state using UpdateUserRoleRequest and UpdateUserStatusRequest.
- **POST** /api/admin/users/create-agent – allow admins to create new agent accounts with CreateAgentRequest.
- **GET** /api/admin/users/agents – subset view of only agents.



- Integrated admin dashboard statistics and recent activity:
 - **GET** /api/admin/dashboard/summary – overall KPIs consumed by AdminDashboardPage.
 - **GET** /api/admin/stats/users/daily, /api/admin/stats/appointments/daily, /api/admin/stats/plans/daily – generic stats series.
 - Wrapper endpoints used by React: /api/admin/users/stats/daily and /api/admin/users/appointments/daily for charts.
 - **GET** /api/admin/users/appointments/recent and GET /api/admin/users/latest for “recent appointments” and “latest users” tables
- Completed admin policy management endpoints in AdminPolicyController:
 - **GET** /api/admin/policies/plans – admin view of all plans.
 - **POST** /api/admin/policies/plans, /api/admin/policies/plans/{id}, **DELETE** /api/admin/policies/plans/{id} – full CRUD over plans.
 - **GET** /api/admin/policies/stats/categories and GET /api/admin/policies/stats/usage - data for policy charts in AdminChartsPanel

Nov 21, 2025

Module #5: AI Integration



Topic

AI Leverage

Data Collection & Processing

Automates extraction from PDFs, voice, and IoT devices.

Underwriting Automation

Uses predictive models for dynamic risk-based pricing.

Claims Processing

CV + RPA for fast, accurate claim settlements.

Customer Interaction

Conversational AI and sentiment analysis.

Risk Management

Predictive analytics and anomaly detection.

Bias Mitigation

Fairness-aware training and explainable models.

Telematics

IoT data from vehicles for personalized premiums.

Continuous Learning

Feedback loops for model retraining.

Understanding and Scope

Well the major challenges faced by insurance companies are

Trust Deficit – Customers often don't trust insurers because of opaque pricing, slow claims, and perceived bias in payouts.

Policy Comprehension – Insurance contracts are full of jargon and fine print that most of them can't decode.

Claims & Renewal Pain Points – Lengthy claim verification, fraud checks, and manual renewals frustrate users and waste company time.

Now using let's see how we can overcome the challenges in this industry

Trust Deficit - Deploying AI models to show why a claim was approved or denied along with sentiment analysis on customer feedback. **(Future Plans)**

Policy Understanding - Using clearly & user friendly UI and recommendation bots that explain policies in plain language and suggest best-fit options.

Claims & Renewals - By Tracking policy dates and customer data to auto-renew or alert users with one-click options, removing all manual follow-ups.

Reference Link

Well there are many blogs and posts regarding use cases and challenges for this topic , these are some blogs whose contents are easier to grasp .

[Link 1](#) [Link 2](#)

Conclusion and Future

Week 6 transitioned InsurAI from isolated backend modules to a cohesive full-stack application. All essential REST endpoints were documented and connected to their corresponding React pages and the Admin/Plans features were brought online with real data. The platform now supports end-to-end insurance workflows for customers, agents, and admins, and is ready for further UX polishing, My Plans implementation, and production-level hardening.

- Add richer validations, error messages, and edge-case handling on the frontend for all critical forms
- Integrating payment gateway API to make it a fully functional software